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Predicting Logistics Delivery Delay Risk Using a Hybrid SMOTE and PSO-Based Optimization Approach with Extra Trees Classifier

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Abstract

Maintaining logistics reliability in Indonesia presents persistent challenges regarding delivery speed and Service Level Agreement (SLA) compliance, especially during peak periods. Conventional logistics planning often lacks the necessary foresight to anticipate delivery delays, necessitating advanced data-driven risk prediction systems. This study introduces a robust hybrid machine learning framework integrating the Synthetic Minority Over-sampling Technique (SMOTE) a technique for handling imbalanced data with Particle Swarm Optimization (PSO) a metaheuristic optimization method to predict delivery delay risks effectively. To address the severe class imbalance in delay data, eight algorithms were rigorously evaluated through stratified cross-validation, identifying the Extra Trees Classifier as the optimal performer. The model was further refined using a PSO-based evolutionary search mechanism (n=167, d=9), achieving an Accuracy of 0.9048, a ROC-AUC of 0.9595, and an F1-Score of 0.8963, significantly outperforming baseline and conventionally tuned configurations. Interpretability analysis using Mean Decrease in Impurity (MDI) a feature importance technique highlighted Posting Hour and Service Type as the primary temporal and operational risk drivers. The final framework generates probabilistic risk scores ranging from 0 to 100 percent, with a peak observed value of 66.38, facilitating actionable early warnings. Validation on real-world datasets confirms the system's efficacy in proactive risk identification and targeted intervention, demonstrating robust suitability for industrial deployment. This study advances machine learning in logistics through an interpretable, optimization-driven framework, with limitations related to the dataset, and future research may incorporate additional factors to improve performance.

Keywords *: logistics delivery risk, extra trees, machine learning, pso, smote.*

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1. INTRODUCTION

Logistics reliability is the cornerstone of supply chain management, directly impacting customer satisfaction and economic stability [1], [2]. However, accurately estimating delivery delay risks remains challenging due to complex, non-linear relationships between fluctuating volumes and operational constraints [3], [4]. These challenges are further compounded by high-dimensional logistics data and extreme class imbalance, where variations in posting hours and service types significantly influence Service Level Agreement (SLA) breach probabilities [5], [6]. Despite these advancements, existing logistics studies still face limitations in capturing real-time operational variability and integrating risk prediction into actionable decision-support systems for SLA management [2], [6]. Recent studies have also emphasized the growing importance of data-driven approaches in logistics risk prediction, particularly in handling uncertainty and operational variability [6], [7]Consequently, interdependent factors necessitate robust analytical approaches capable of learning from imbalanced data without majority-class bias .

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Machine learning methods offer powerful capabilities for handling such complexity, particularly when comparing various algorithms to find the most effective strategy. In this study, eight classifiers are evaluated: Logistic Regression, Decision Trees, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, K-Nearest Neighbors, and Support Vector Machines. Among these, Extra Trees is identified as a primary candidate for high-dimensional logistics tasks due to its randomized splitting mechanisms [8], [9], [10]. To maximize performance, Particle Swarm Optimization (PSO) is employed as a superior hyperparameter tuning alternative to computationally expensive Grid Search or stochastic Random Search [11], [12], [13], [14]. Advances in metaheuristic optimization have further demonstrated their effectiveness in improving model generalization and avoiding local optima in complex datasets[15]. To ensure a rigorous assessment of imbalanced data, performance is evaluated using multi-dimensional metrics such as F1-Score and ROC-AUC, which balance overall correctness with the specific ability to detect minority class failures[16], [17], [18].

Addressing data sparsity using SMOTE is essential to mitigate majority class bias; however, its integration with PSO-optimized Extra Trees models remains limited in logistics delay prediction studies[6], [7], [19]. Several studies have explored SMOTE-based approaches for imbalanced learning; their integration with ensemble models and optimization techniques is still underexplored[20], [21]. In addition, many existing approaches prioritize predictive accuracy without sufficient emphasis on interpretability and operational usability. Motivated by this gap, this study integrates SMOTE, Extra Trees, and Particle Swarm Optimization (PSO) to improve minority delay detection and model robustness [20], [21], [22]. Feature importance analysis using Mean Decrease in Impurity (MDI) is employed to identify key delay drivers and enhance interpretability [23], [24], [25]. The proposed framework produces both binary SLA risk classifications and probabilistic risk scores (0–100%), enabling practical early-warning support for proactive SLA breach prevention. This combination is expected to outperform existing approaches by effectively addressing class imbalance, optimizing model performance, and maintaining interpretability within a unified framework.

2. METHOD

The methodological framework (Figure 1) consists of five stages: (1) data preparation and SMOTE resampling [19]; (2) modeling eight classifiers (LR, DT, RF, KNN, SVM, ET, GB, AB) with PSO,GS,RS hyperparameter optimization; (3) champion model identification; (4) MDI-based feature importance analysis; and (5) performance evaluation and interpretation. The study predicts binary delivery status (on-time vs. delayed) using operational factors like shipment type, SLA, and temporal features. Pre-processing includes One-Hot Encoding and feature scaling, while SMOTE addresses class imbalance. Hyperparameter tuning via PSO optimizes critical parameters such as *n_estimators* and *max_depth*. Model performance using F1-Score, ROC-AUC, and Accuracy, with MDI identifying the primary risk drivers [16], [23].

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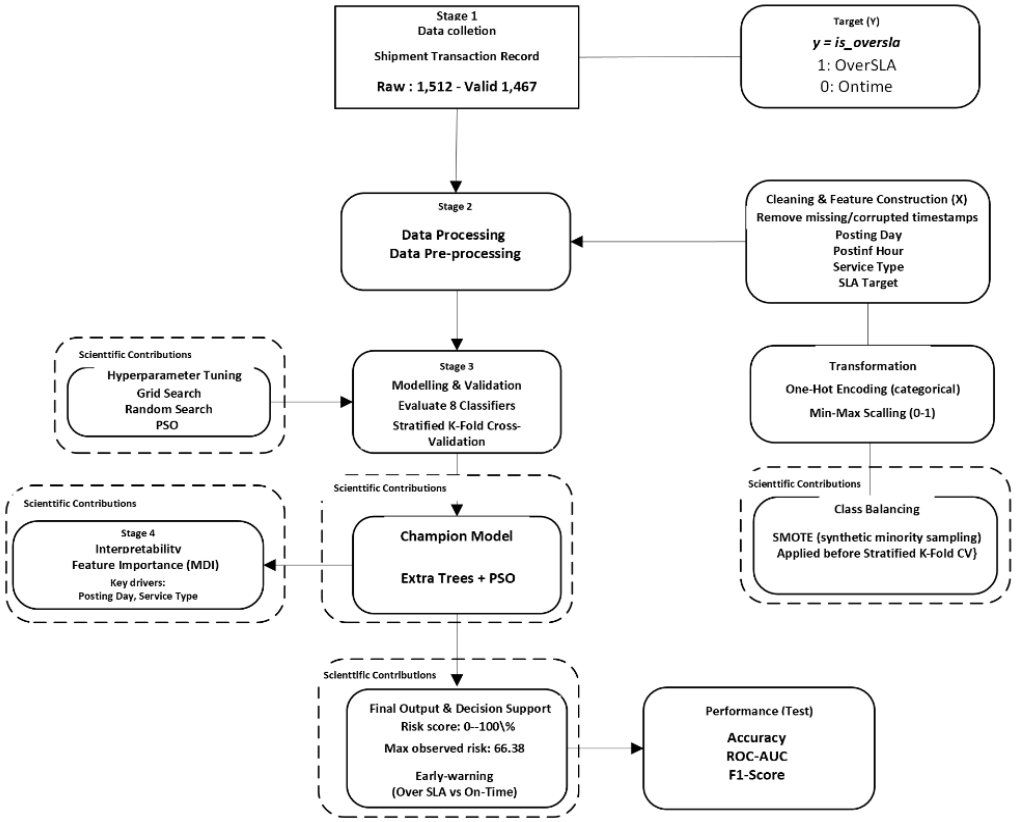


Figure 1. Methodology

2.1. Data Collection

This study utilizes 1,512 transaction records from an Indonesian logistics provider's Asia-Africa Distribution Center, collected between October 29 and November 8, 2025. The raw data are structured into three primary categories: identity factors (e.g., service type), time factors (e.g., shipment date and SLA deadline), and tracking factors (e.g., final status and milestone logs). Receipt numbers serve solely as anonymized indices, and records with incomplete critical timestamps were removed to ensure data integrity. Selected features for modeling include *service_level* and temporal patterns (*posting_hour*, *posting_day*). The binary target, *is_oversla*, classifies shipments as high risk (1) if they exceed the *sla_deadline*, while compliant shipments are labeled on-time (0). Table 1 summarizes these variables including categorical features such as *service_level* and *posting_day*, as well as numerical features like *sla_target* and *posting_hour*, along with their statistical distributions (median, IQR, range, and p-value).

Tabel 1. Summary of research variables.

Variable	Type	Description	Median	IRQ	Min-Max	P-Value
service_level	Categorical	service type	-	-	-	-
posting_day	Categorical	shipment	-	-	-	-
sla_target	Numerical	target (days).	2.00	1.00	[0, 15]	<0.001
posting_hour	Numerical	Proces hour.	11.00	14.00	[0, 23]	<0.001

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2.2. Data Processing and Pre-Processing

Data preprocessing ensures statistical consistency and predictive integrity. The process begins with rigorous data cleaning, excluding records with missing or corrupted timestamps to maintain quality. A key step is constructing the binary target variable y, representing delivery risk status based on SLA violation[6]. The target is defined in Eq. (1):

$$y = is_{oversla} = \begin{cases} 1 \text{ (High Risk) if } T_{actual} > T_{deadline} \\ 0 \text{ (On-Time) if } T_{actual} \leq T_{deadline} \end{cases} \quad (1)$$

Here, T_{actual} and $T_{deadline}$ represent actual delivery time and the SLA threshold. To optimize feature representation, categorical variables are transformed via One-Hot Encoding, while numerical features are normalized using Min-Max scaling to a range of [0, 1][25].

Exploratory analysis revealed a class imbalance, with on-time shipments outnumbering high-risk cases to mitigate this, the SMOTE is applied[20]. generates synthetic minority samples through vector interpolation, as defined in Eq. (2):

$$S_{syn} = S_f + r \cdot (S_{kNN} - S_f) \quad (2)$$

In this formulation, S_{syn} is the synthetic instance, S_f is the focal minority sample, S_{kNN} is a randomly selected k-nearest neighbor, and r controls interpolation[21]. By generating these synthetic samples, the dataset becomes balanced, preventing the classifier from becoming biased toward the majority of on-time shipments. This ensures the model captures subtle delay patterns, resulting in more reliable risk assessments for high-risk shipments.

2.3. Classification Modeling And Evaluation

Modeling begins with an 80:20 split and SMOTE integration to resolve class imbalance[17]. In simple terms, SMOTE addresses class imbalance by generating synthetic samples for the minority class through interpolation between existing data points, thereby creating a more balanced training distribution. Eight algorithms (LR, DT, RF, KNN, SVM, GB, AB, and ET) are evaluated using Stratified K-Fold Cross-Validation[8]. ET is selected as the primary architecture unlike standard bagging[1]. ET constructs ensembles by selecting split points entirely at random across the full training set to minimize variance and enhance robustness. The final prediction ŷ is derived through majority formalized in Eq. (3):

$$y(x) = \backslash text(mode)\{ h_{1(x)}, h_{2(x)}, \dots, h_{M(x)} \} \quad (3)$$

Within this framework, {y}(x) represents the final consensus class, while M signifies the total number of trees. Tree construction relies on impurity measures to evaluate the quality of random splits. Two fundamental metrics, Gini impurity in Eq. (4) and entropy in Eq. (5), quantify node disorder:

$$(Gini\ Impurity) = \sum_{j=1}^{O} f_j(1 - f_j) \quad (4)$$

$$(Entropy) = \sum_{j=1}^{O} -f_j log(f_j) \quad (5)$$

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Gini impurity measures misclassification probability by aggregating class variance (fj) across O target classes via the $\sum_{j=1}^{O}$ operator; a value of 0 indicates perfect purity. Similarly, entropy quantifies uncertainty by accumulating information content $log(f_j)$ across O classes. Minimizing these metrics maximizes information gain for optimal node splits between 'On-Time' and 'High-Risk' shipments. Model performance is evaluated using accuracy, macro-average f1-score, and ROC-AUC[18].

2.4. Hyperparameter Optimization via PSO

Hyperparameter tuning follows a three-stage framework comprising GS, RS, and PSO[14], [26]. Conceptually, PSO operates like a coordinated search process, where multiple candidate solutions iteratively move within the search space by adapting based on both their own performance and the best solution identified by the swarm. PSO is utilized as the primary engine to maximize the model's F1-Score. PSO navigates the search space by balancing individual best experiences $\mathrm{p_{best}}$ and collective swarm intelligence $\mathrm{g_{best}}$[12]. The search dynamics are defined by the velocity and position updates in Equations 6 and 7 :

$$v_{\{i\}(t+1)} = \omega \cdot v_{\{i\}(t)} + c_{\{1\}} r_{\{1\}\left(p_{\{best,i\}} - x_{\{i\}(t)}\right)} + c_{\{2\}} r_{\{2\}\left(g_{\{best\}} - x_{\{i\}(t)}\right)} \quad (6)$$

$$x_{\{i\}(t+1)} = x_{\{i\}(t)} + v_{\{i\}(t+1)} \quad (7)$$

In Equation 6, $\mathbf{v_i(t+1)}$ signifies momentum, where inertia weight $\boldsymbol{\omega}$ balances exploration and exploitation by regulating $\mathbf{v_i(t)}$. Acceleration coefficients $\mathbf{c_1}$ and $\mathbf{c_2}$, guided by random variables $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$, weigh cognitive and social influences to bypass local optima. Equation 7 then updates the position $\boldsymbol{x_i(t+1)}$ until the swarm identifies the optimal number of estimators and maximum depth. This iterative process secures the global optimum for the Macro-Average F1-Score, addressing the non-linear class boundaries inherent in logistics datasets.

2.5. Feature Important Analysis : Mean Decrease in Impuriry

Feature relevance is evaluated using MDI. MDI identifies operational drivers by quantifying the weighted Gini impurity reduction attributed to each feature across the ensemble[23]. The importance of feature k within tree T is defined in Eq. (8):

$$MDI(k,T) = \sum_{t \in I(T), s(t)=k} \left(\frac{n(t)}{n}\right) \Delta I(t) \quad (8)$$

Impurity reduction $\Delta\, \mathcal{I}\,(t)$ is aggregated across nodes t where k is the split criterion, weighted by the sample ratio $n(t)/n$. To address fragmentation from One-Hot encoding, post-hoc aggregation sums binary component scores into unified operational dimensions (e.g., service level).

3. RESULT

First Stage refined 1,512 records into 1,467 observations (97.02% retention), minimizing noise for robust modeling [27]. The feature set includes critical variables such as service level, sla target, posting hour and posting day. The binary target $y_{is_oversla}$ classifies shipments as High Risk $y = 1$ for Service Level Agreement breaches or On-Time $y = 0$, aligning with standard risk modeling

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formulations. To ensure integrity, One-Hot Encoding prevented ordinal bias in categorical data, while Min-Max Scaling normalized numerical features to [0,1] to prevent scale dominance. To rectify severe class imbalance, the SMOTE was employed. As shown in (Figure 2), this method uses feature-space interpolation to balance the dataset, enabling the model to better identify minority delay events [20].

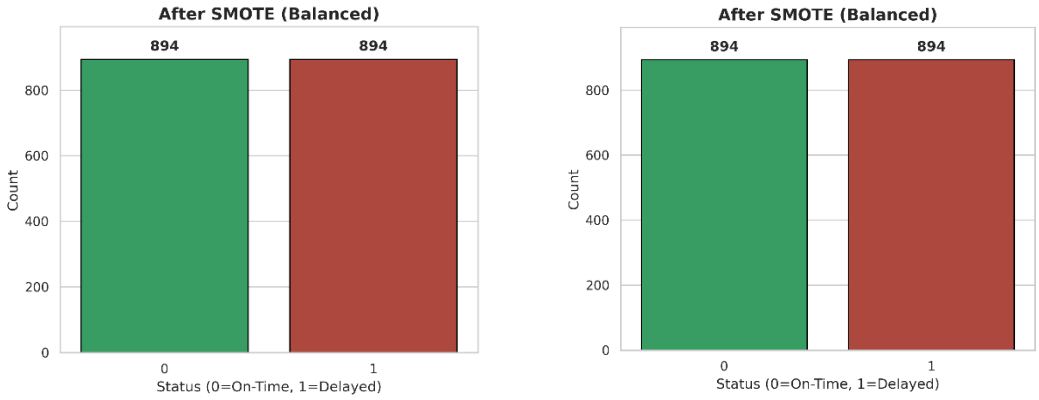


Figure 2. Class distribution: (a) raw imbalance and (b) balanced via SMOTE

Furthermore, temporal heat map analysis (Figure 3) shows that delivery delays are concentrated during late cut-off hours, indicating a clear time-dependent risk pattern. This finding further strengthens the inclusion of posting hour as a key predictive feature and emphasizes the critical role of temporal exploratory analysis in enhancing logistics delay risk modeling and operational decision-making [6]

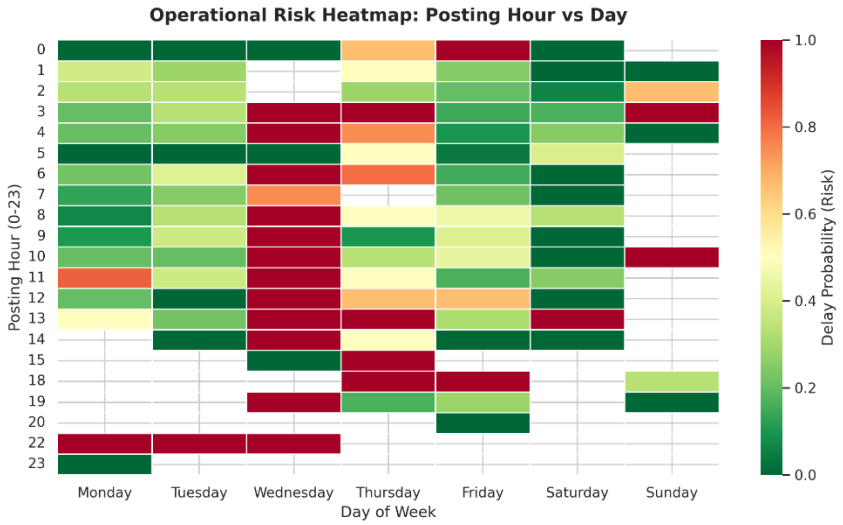


Figure 3. Risk heatmap: Delay clusters in late posting hours

Second Stage evaluates eight classifiers using Stratified K-Fold CV across three tuning strategies: GS, RS, and PSO[14], [26]. Random Search showed no meaningful improvement over the baseline model and the lowest ROC-AUC (0.9520), indicating limited convergence capability[13]. Among the candidates, Extra Trees consistently demonstrated superior performance.

While Grid Search achieved a marginally higher ROC-AUC (0.9602 vs. 0.9595), the difference was statistically negligible. Ultimately, the PSO-optimized Extra Trees achieved the highest F1-Score (0.8963). This configuration was selected as the champion model for its optimal balance of precision and false negative mitigation [28]. Figure 4 illustrates the performance comparison between baseline

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models and models optimized using various hyperparameter tuning methods, including Grid Search, Random Search, and PSO.

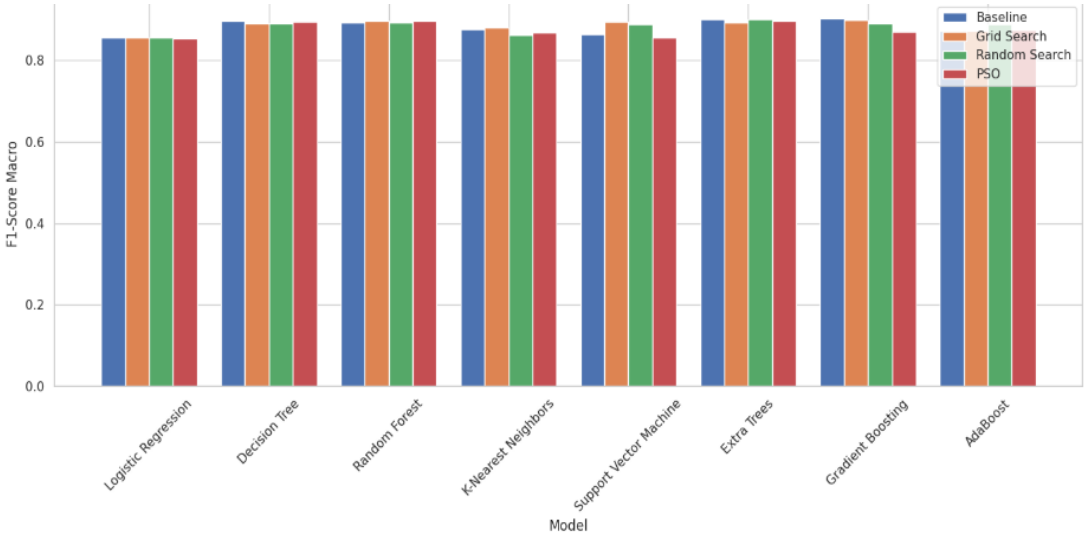


Figure 4 Comparison of baseline models

Table 2 presents the performance comparison of multiple classification models using different optimization methods (Grid Search, Random Search, and PSO), evaluated based on Accuracy, ROC-AUC, and F1-Score.

Tabel 2. Comprehensive performance grouped by Optimization Method

Model	Method	Accuracy	ROC-AUC	F1-Score
LR	Grid Seach	0.8673	0.9313	0.8559
DT	Grid Seach	0.8435	0.8524	0.8428
RF	Grid Seach	0.8980	0.9473	0.8873
ET	Grid Seach	0.9014	0.9602	0.8928
GB	Grid Seach	0.9082	0.9542	0.8998
AB	Grid Seach	0.8742	0.9023	0.8597
KNN	Grid Seach	0.8333	0.9080	0.8145
SVM	Grid Seach	0.8776	0.9389	0.8602
LR	Random Search	0.8673	0.9362	0.8559
DT		0.8559	0.8594	0.8306
RF	Random Search	0.8946	0.9459	0.8837
ET	Random Search	0.9082	0.9520	0.8998
GB		0.9082	0.9516	0.8998
AB		0.8742	0.9137	0.8597

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KNN	Random Search	0.8333	0.9080	0.8145
SVM	Random Search	0.8776	0.9377	0.8602
LR	PSO	0.8639	0.9439	0.8525
DT	PSO	0.8571	0.8407	0.8571
RF	PSO	0.9014	0.9513	0.8927
ET	PSO	0.9048	0.9595	0.8963
GB	PSO	0.9116	0.9526	0.9043
AB	PSO	0.8742	0.9105	0.8597
KNN	PSO	0.8435	0.9127	0.8277
SVM	PSO	0.8810	0.9438	0.8655

The third stage focuses on fine-tuning the Extra Trees champion model through a comparative analysis of optimization strategies[13]. While Grid Search provided a stable baseline, the PSO approach demonstrated superior adaptive capability[15]. The heuristic nature of PSO enabled effective convergence toward an optimal configuration (*n_estimators* = 167, *max_depth* = 9), achieving the highest F1-Score of 0.8963. This confirms PSO as a robust strategy for maximizing minority class detection, effectively balancing model complexity with predictive sensitivity[28].

The efficiency of this phase is evidenced by the algorithm's learning behavior. As shown in (Figure 5), the PSO convergence curve reveals rapid F1-Score improvement, indicating its effectiveness in escaping local optima to identify global parameters[13]. Furthermore, (Figure 6) highlights the significant performance uplift achieved by PSO over traditional Grid Search, specifically in enhancing classification precision[15].

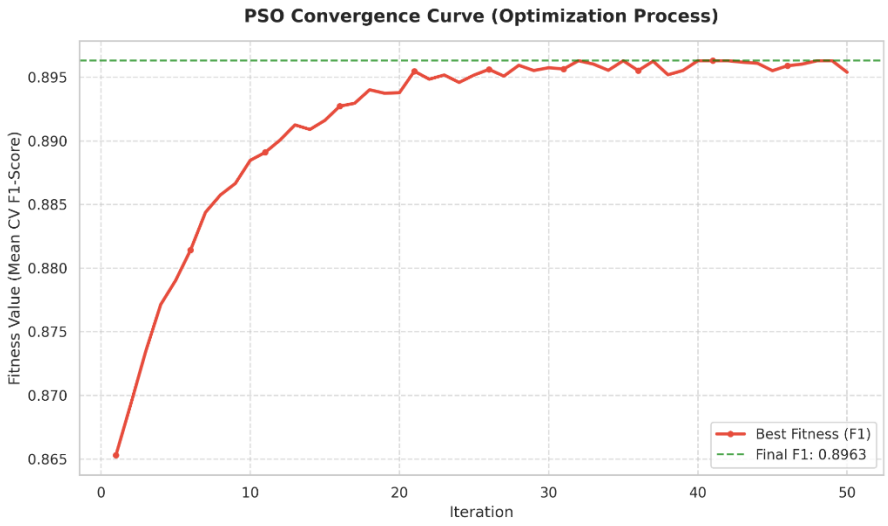


Figure 5. PSO convergence: Rapid F1-Score improvement.

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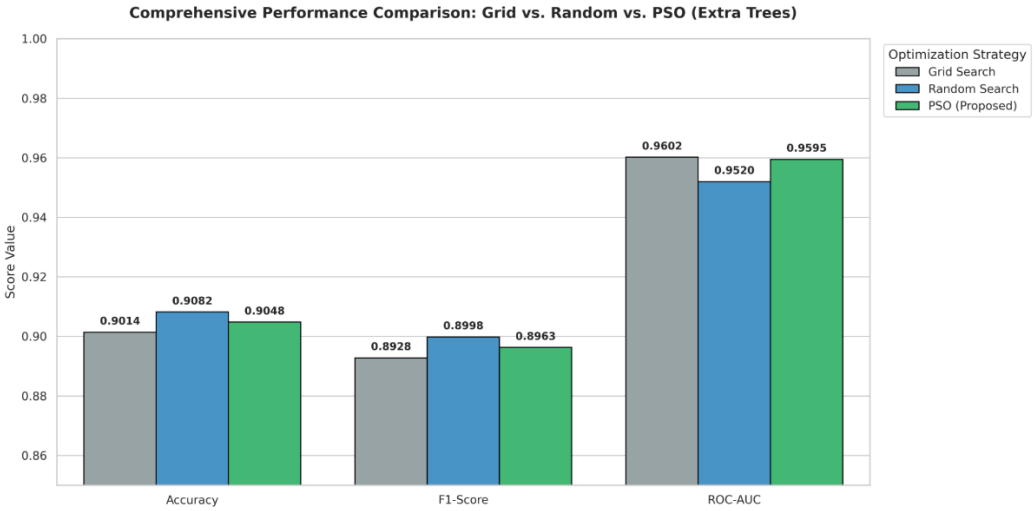


Figure 6.ET Multi-metric results by tuning strategy.

The fourth stage uses Mean Decrease in Impurity (MDI) to interpret the PSO-optimized ET model[10]. As shown in Figure 7, Posting Day (51.4%) is the primary risk driver linked to weekly patterns, followed by Service Type (25.9%), SLA Target (14.6%), and Posting Hour (8.2%)[28]. This feature importance hierarchy indicates that the model effectively captures key logistics dynamics, highlighting Posting Day as a critical factor for proactive operational monitoring and preemptive management intervention[10], [28]

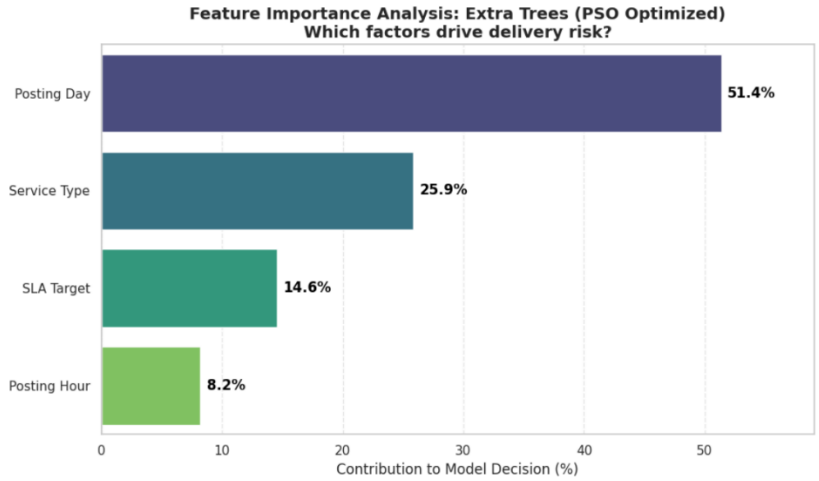


Figure 7. Feature Importance hierarchy based on MDI

The final stage validates the model through real-world risk-scoring and prescriptive decision support [28]. The model transforms predictions into an actionable dashboard, classifying shipments into Over SLA and On-Time categories. As shown in the operational analysis, shipments posted at 19:00 carry a significant risk score (66-67%), triggering Priority Monitoring alerts[28]. In contrast, shipments at 07:00 show lower risk (38-39%) and follow Standard Process workflows. Table 3 presents the operational risk synthesis, showing predicted risk scores, classification results, and corresponding action recommendations for each shipment.

Tabel 3. Operational Risk Synthesis: Predictive Scores and Prescriptive Action Plans

Receipt	Service	PH	Risk%	Prediction	Action	Validation
P2510300002771	PKH	19	66.38	Over SLA	PM	Match
P2510300002810	PKH	19	67.24	Over SLA	PM	Match
P2511030090432	PKH	07	39.76	On Time	SP	Match

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P2511030096694	PKH	07	38.66	On Time	SP	Match

This granular precision validates the impact of PSO-optimization, demonstrating superior sensitivity in identifying potential delivery failures[15]. The consistency between predicted risk levels and observed outcomes across peak and off-peak operational periods indicates that the proposed framework provides a reliable, decision-oriented tool for proactive logistics risk mitigation[13] These findings provide practical implications for logistics operations by enabling early identification of high-risk shipments and facilitating more informed decision-making as well as efficient resource allocation[2], [6].

4. DISCUSSIONS

The experimental results highlight that severe class imbalance and nonlinear temporal patterns explain why conventional classifiers fail in logistics delay prediction[4], [7]. Unlike prior studies that rely on deep learning or gradient boosting without resampling[25], this research integrates SMOTE to enhance minority class detection. This directly addresses the common research gap where SLA breach events are underrepresented and "underlearned" in postal service datasets. While most literature emphasizes Random Forest or XGBoost[29], this study identifies Extra Trees as a more stable alternative for noisy logistics data[9]. Furthermore, the incorporation of Particle Swarm Optimization (PSO) differentiates this framework from existing approaches that typically rely on limited grid or random search strategies[25]. The PSO-optimized Extra Trees model achieves the highest F1-Score, validating the advantage of metaheuristic optimization in balancing precision and recall a critical requirement for minimizing operational false negatives[29].

Using MDI, the analysis identifies Posting Day and Service Type as the primary drivers of delivery risk[24]. This extends prior interpretability studies by explicitly linking feature relevance to SLA-oriented classification. By transforming these outputs into a probabilistic risk-scoring system, the framework enables targeted operational interventions. Ultimately, the SMOTE–PSO–Extra Trees framework systematically bridges key gaps including imbalance-aware modeling, metaheuristic adoption, and actionable risk-scoring positioning it as an industry-ready solution for proactive logistics risk mitigation. This study contributes to computer science by demonstrating an effective integration of SMOTE, PSO, and Extra Trees to improve prediction performance and interpretability in imbalanced datasets. These findings are consistent with broader logistics studies that emphasize the importance of data-driven risk prediction in improving delivery reliability and operational efficiency within supply chain systems [6].

5. CONCLUSION

This study demonstrates that the PSO-optimized ET framework provides a robust and interpretable solution for predicting logistics delivery risks. The champion model (n=167, d=9) successfully captures nonlinear relationships within imbalanced operational data, achieving an Accuracy of 0.9048, ROC-AUC of 0.9595, and F1-Score of 0.8963. By integrating SMOTE to resolve class imbalance and MDI analysis to deconstruct model logic, the research identifies Posting Day and Service Type as the primary determinants of delivery risk. These findings enable proactive SLA management and targeted resource allocation. Ultimately, this framework offers an industry-ready foundation for data-driven logistics planning, with future potential for real-time monitoring and the inclusion of external variables like traffic and weather conditions. This study also contributes to computer science by providing a hybrid machine learning framework that improves predictive accuracy and

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interpretability for imbalanced data problems. Despite its strong performance, this study is limited to a specific dataset and operational context, which may affect its generalizability to other logistics environments. From a practical perspective, the proposed framework can be implemented as a decision-support tool for early risk detection, enabling logistics companies to improve SLA monitoring and optimize resource allocation. Future research should focus on integrating real-time external data such as traffic conditions and weather information, as well as evaluating the model in diverse logistics scenarios to enhance robustness and scalability.

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